

Blockchain Technology

this session will give you a complete idea of how a blockchain system works and support. The session will discuss how bitcoins can help solve various issues and how bitcoin transactions take place. We will also talk about various blockchain concepts and features, and see where blockchain is being used apart from banking. In the latter half of this session, we'll be looking at a banking use case demo where we try to create an alternative banking system. The session will be held on Monday at 8 pm.

The challenges faced with the current existing banking system include transaction fees, double spending, financial crisis, financial depression, and net banking fraud. A parallel system is required to solve these challenges. Let's first talk about the transaction fee. The current existing system has a huge amount of transaction fees which is due to be fixed. Most banking and financial organizations depend on these kinds of fees as a source of income and revenue. In the year 2015, three organizations, i.e. CPM, Irving Bank of America, and Wells Fargo, made close to 6 billion just from ATM and overdraft transaction fees.

The digital signature attached to digital transactions can be falsified and copied, making double spending quite possible; something that bitcoin solves the issue of. The bitcoin system is decentralized and distributed among everyone who is part of it, ensuring that power is not held by a single authority. Bitcoin is a public ledger, making it easy to have transaction details. The blockchain makes it completely secure to store all transaction details and ensures that none of it gets manipulated as well. The anonymity that blockchain gives is also beneficial.

Bitcoin has a huge number of transactions, but the blockchain makes it easy to store all this transaction detail and at the same time ensures that none of it gets manipulated as well. In our upcoming sessions, we will discuss how the value of bitcoins keeps changing and how blockchain makes it possible. Most financial organizations with a digital platform face the challenge of double spending through blockchain, but this is not possible through blockchain as even if one tries to modify a false transaction, it gets rejected due to how the system is structured. The system was structured to prevent double spending and is commonly used today in payment platforms. Bitcoin was valued at \$1,100 in January 2011 but today, it stands close to \$2,800 due to the increasing trust in the system and more people investing in it.

There are many security features in this system and this session will discuss how to purchase cryptocurrency, how it benefits you and how to secure your wallet. Chum Shaker asks how to purchase a cryptocurrency or bitcoins, and it is quite easy to do so by purchasing a bitcoin.

Tipica asks if there is an option for no transaction fees in Bitcoin, however, Bitcoin is the first decentralized digital currency introduced in 2009 with only 21 million available. The community keeps upgrading algorithms and cryptographic technologies to ensure the system is not exhausted even with the increase in complexity. Every block is grouped and stored in the blockchain where it becomes a permanent part of the chain and the chain continues to grow every ten minutes with new transactions.

Polly says that anyone can use bitcoin and that the blockchain system can't be directly hacked. Even if someone tries to falsify or fake a transaction or detail, it won't be accepted by the ledger. Since the ledger is distributed throughout the network, everyone has a valid copy of it. Although applications built on top of the blockchain have been hacked before, the blockchain system itself has never been hacked.

Polly notes that there have been incidents where applications built upon the blockchain have been hacked, but we'll talk more about that in upcoming slides. She also explains that a group of people were shown how a bitcoin transaction takes place and they will discuss the protocols and security features involved. They'll mainly focus on the unspent transaction outputs and how transactions are validated.

Lisa and Robert are going to check the transaction history of chips to ensure James has enough funds to transfer bitcoins to Kevin. Once the transaction is verified and added to a block, the money is taken from James's account and put into Kevin's bitcoin wallet to avoid double spending. It usually takes a minute to validate a block, but it is suggested to wait about 30 minutes to confirm a transaction.

Patients often ask me if Lisa and Robert actually sit down and validate their transactions, but it's not easy to validate a transaction as it requires a lot of resources. In today's world, it is highly impossible for a single person or entity to mine a complete block, so it requires a lot of investment. Why do you believe James, Lisa, or Robert do that? We will discuss it in the next session where I will show you how it is impossible to do a double spending challenge.

Since both of them are trying to confirm this transaction, it can take them different times to confirm it. But even then, there is a solution to that. What

happens is the message gets transmitted across multiple people in the network. The person who solves the next block puts in the transaction confirmation from either Lisa or Robert, and whichever transaction they have accepted will be accepted and the other transaction will get rejected. The transaction charges are completely neglected here. Even if the miners who validate the transaction and complete the next block of the blockchain get a 12.5 bitcoin as an incentive for the creation of the new block, mining requires a lot of investment and it's not easy for a single person to mine the resource today.

A lot of miners come together and form something called a pool. These people invest all the resources that they have to solve the pool together, and based on how many resources they have, the pool reward gets distributed accordingly. The complexity involved in solving a block keeps growing day by day, and the bitcoin system does not need to ensure that the bitcoin system doesn't get depleted. There is a lot more people mining today when compared to six or eight months ago, and there are only 21 million bitcoins that can be mined so the system makes it difficult to compute the resource.

Once a transaction gets verified and validated, it becomes a part of the new block which gets added to the existing blockchain. It is again a publicly distributed database where you're holding encrypted ledgers of everyone involved in the bitcoin system. The number of transactions at this point could be all the recent transactions or only those that have been completed as well. This depends on the total number of transactions that get verified and validated in the time that a new block gets created.

Each block, once it's completed, gets added to the blockchain, and the next set of transactions gets added onto this block, and in turn, it gets added to the next block, and it keeps going. As I mentioned earlier, once a transaction gets completed, the number of transactions that have been completed gets added, and the next block gets added to the blockchain, and it keeps going.